



Neural Control and Coordination

The functions of the **organs/organ systems** in our body **must be coordinated to maintain homeostasis**.

- **Coordination** - the process through which **two or more organs interact and complement the functions of one another**.

Worked example - physical exercise. When we do physical exercises the **energy demand is increased** for maintaining an increased muscular activity, and the **supply of oxygen is also increased**. That increased oxygen supply necessitates an increase in the **rate of respiration, heart beat and increased blood flow via blood vessels**. When physical exercise is stopped, the activities of **nerves, lungs, heart and kidney** gradually return to their **normal conditions**. Thus the functions of **muscles, lungs, heart, blood vessels, kidney** and other organs are coordinated while performing physical exercises.

System	How it coordinates	Nature of the link
Neural system	Provides an organised network of point-to-point connections	For a quick coordination
Endocrine system	Provides chemical integration	Through hormones

In our body the **neural system and the endocrine system jointly** coordinate and integrate **all** the activities of the organs so that they function in a **synchronised fashion**.

- ① **[NOTE] Scope of this chapter.** You will learn about the **neural system of human and the mechanisms of neural coordination - the transmission of nerve impulse and impulse conduction across a synapse**.

18.1 Neural System

- **Neurons** - the neural system of **all** animals is composed of highly specialised cells called neurons, which can **detect, receive and transmit** different kinds of **stimuli**.

Animal group	Level of neural organisation
Lower invertebrates	Neural organisation is very simple .
<i>Hydra</i>	Composed of a network of neurons .
Insects	The neural system is better organised : a brain is present along with a number of ganglia and neural tissues .
Vertebrates	Have a more developed neural system.

18.2 Human Neural System

The human neural system is divided into **two parts**: (i) the **central neural system (CNS)** and (ii) the **peripheral neural system (PNS)**.

- **CNS** - includes the **brain** and the **spinal cord**, and is the **site of information processing and control**.
- **PNS** - comprises **all the nerves of the body associated with the CNS** (brain and spinal cord).

The nerve fibres of the PNS are of two types: (a) **afferent fibres** and (b) **efferent fibres**.

Fibre type	Direction of the impulse	What it carries
Afferent nerve fibres	From tissues/organs to the CNS	Sensory input arriving at the CNS
Efferent fibres	From the CNS to the concerned peripheral tissues/organs	Regulatory impulses

The PNS is divided into two divisions called the **somatic neural system** and the **autonomic neural system**.

Division of the PNS	Relays impulses from the CNS to
Somatic neural system	Skeletal muscles
Autonomic neural system	The involuntary organs and smooth muscles of the body

- The autonomic neural system is **further classified** into the **sympathetic neural system** and the **parasympathetic neural system**.
- **Visceral nervous system** - the part of the **peripheral** nervous system that comprises the whole complex of **nerves, fibres, ganglia and plexuses** by which impulses travel **from the central nervous system to the viscera** and **from the viscera to the central nervous system**.

① **[NOTE]** Exercise gap closed here (Exercise 10f: cranial nerves vs spinal nerves). NCERT's own text names **spinal and cranial nerves** only once, as the place where myelinated nerve fibres are found (18.3), and never defines either. Both are **PNS** nerves and they are told apart by **where they leave the CNS: cranial nerves arise from the brain**, while **spinal nerves arise from the spinal cord**. Each carries afferent fibres, efferent fibres, or both. Beyond this chapter's own sentences - kept because Exercise 10(f) demands the distinction (Rule 2).

18.3

Neuron as Structural and Functional Unit of Neural System

A **neuron** is a **microscopic** structure composed of **three major parts**, namely, **cell body, dendrites** and **axon** (Figure 18.1).

Part of the neuron	Structure	Function
Cell body	Contains cytoplasm with typical cell organelles and certain granular bodies called Nissl's granules	The metabolic centre of the neuron; impulses arrive here from the dendrites
Dendrites	Short fibres which branch repeatedly and project out of the cell body ; they also contain Nissl's granules	Transmit impulses towards the cell body
Axon	A long fibre , the distal end of which is branched ; each branch terminates as a bulb-like structure called a synaptic knob , which possesses synaptic vesicles containing chemicals called neurotransmitters	Transmits nerve impulses away from the cell body to a synapse or to a neuro-muscular junction

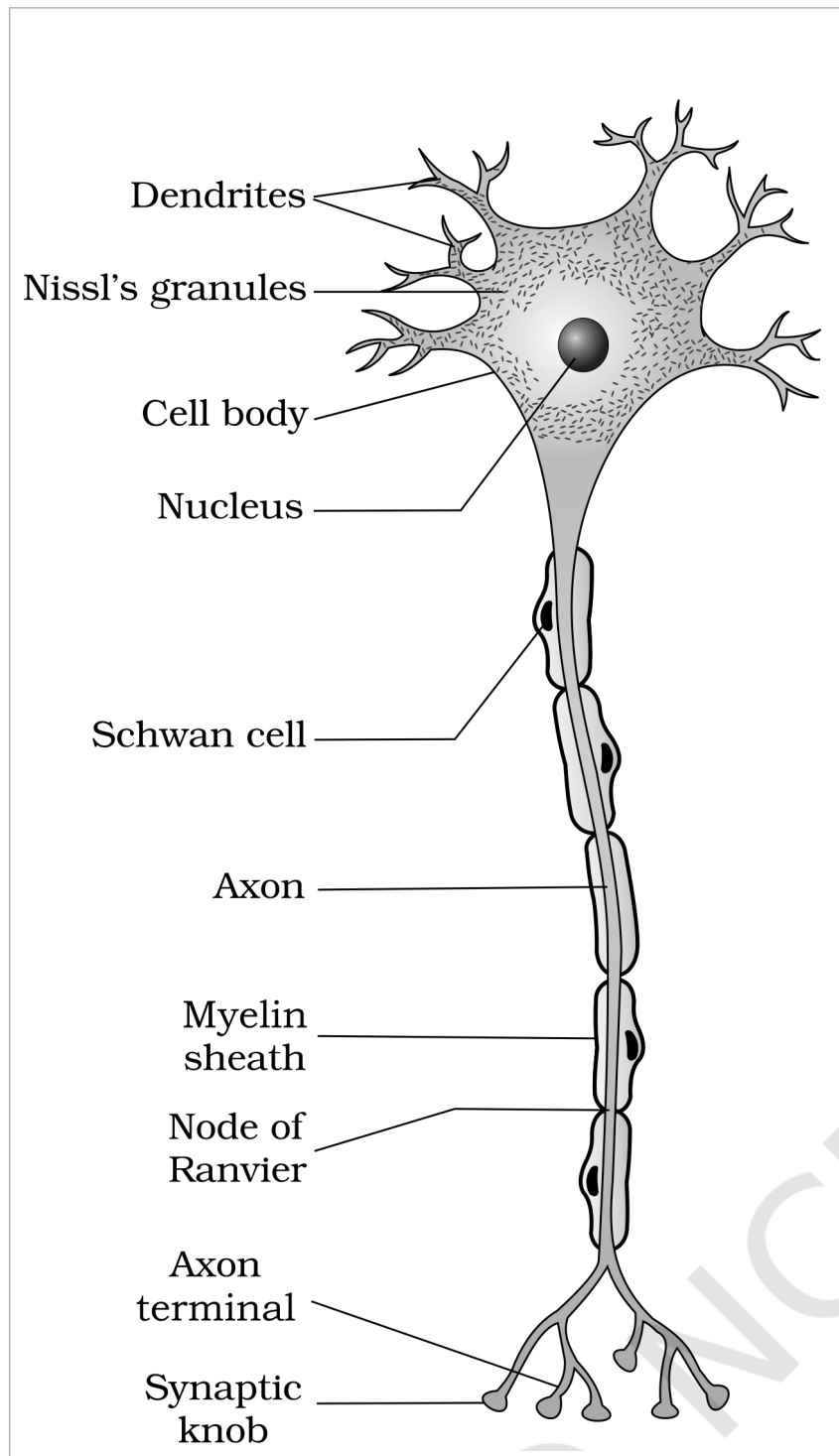


Fig. 18.1 - Structure of a neuron

① [NOTE] Read the plate (Figure 18.1 labels). The diagram marks the **Dendrites**, the **Nissl's granules** inside the **Cell body**, the **Nucleus** within that cell body, the **Schwan cell** (the plate's own spelling of the **Schwann cell**) wrapping the **Axon** as a **Myelin sheath**, a **Node of Ranvier** between two sheaths, and at the far end the **Axon terminal** ending in a **Synaptic knob**.

Based on the number of axon and dendrites, the neurons are divided into **three types**.

Type	Axon and dendrites	Where found
Multipolar	One axon and two or more dendrites	Cerebral cortex
Bipolar	One axon and one dendrite	Retina of eye
Unipolar	Cell body with one axon only	Found usually in the embryonic stage

There are **two types of axons**, namely, **myelinated** and **non-myelinated**.

Feature	Myelinated nerve fibre	Unmyelinated nerve fibre
Schwann cells	Enveloped with Schwann cells , which form a myelin sheath around the axon	Enclosed by a Schwann cell that does not form a myelin sheath around the axon
Gaps in the sheath	The gaps between two adjacent myelin sheaths are called nodes of Ranvier	No myelin sheath, hence no nodes of Ranvier
Where found	Spinal and cranial nerves	Commonly found in autonomous and the somatic neural systems

☆ **[MEMORY AID - not in NCERT]** *Neuron traffic direction: Dendrites Deliver in, Axons Away. Both carry Nissl's granules only on the dendrite side of the cell body.*

18.3.1 Generation and Conduction of Nerve Impulse

Neurons are **excitable cells** because their **membranes are in a polarised state**. *Why is the membrane of a neuron polarised?*

- **Different types of ion channels** are present on the neural membrane, and these ion channels are **selectively permeable to different ions**.
- **At rest** (a neuron **not conducting any impulse**) the axonal membrane is **comparatively more permeable to potassium ions (K^+)** and **nearly impermeable to sodium ions (Na^+)**.
- **Similarly**, the membrane is **impermeable to negatively charged proteins** present in the **axoplasm**.

Compartment	K^+	Na^+	Negatively charged proteins
Axoplasm (inside the axon)	High concentration	Low concentration	High - trapped inside
Fluid outside the axon	Low concentration	High concentration	Not stated by NCERT

These opposed concentrations **form a concentration gradient** across the membrane.

- **Sodium-potassium pump** - the ionic gradients across the resting membrane are maintained by the **active transport** of ions by this pump, which transports **3 Na^+ outwards for 2 K^+ into the cell**.

As a result, the **outer surface** of the axonal membrane possesses a **positive charge** while its **inner surface becomes negatively charged**, and the membrane is therefore **polarised**.

- **Resting potential** - the **electrical potential difference across the resting plasma membrane**.

Now the mechanism of generation of a nerve impulse and its conduction along an axon.

- 1 A **stimulus is applied at a site** (Figure 18.2, e.g., **point A**) on the polarised membrane; the membrane at **site A becomes freely permeable to Na^+** .
- 2 This leads to a **rapid influx of Na^+** , followed by the **reversal of the polarity at that site**: the **outer surface becomes negatively charged** and the **inner side becomes positively charged**. The polarity at site A is thus reversed and hence **depolarised**.
- 3 The electrical potential difference across the plasma membrane at site A is called the **action potential**, which is **in fact termed as a nerve impulse**.
- 4 At sites **immediately ahead** - the axon at **site B** - the membrane still has a **positive charge on the outer surface** and a **negative charge on its inner surface**.
- 5 As a result, a **current flows on the inner surface from site A to site B**; on the **outer surface current flows from site B to site A** (Figure 18.2), **completing the circuit of current flow**.
- 6 Hence the **polarity at the site is reversed** and an **action potential is generated at site B** - i.e. the impulse (action potential) generated at **site A arrives at site B**.
- 7 The sequence is repeated along the length of the axon and consequently the **impulse is conducted**.

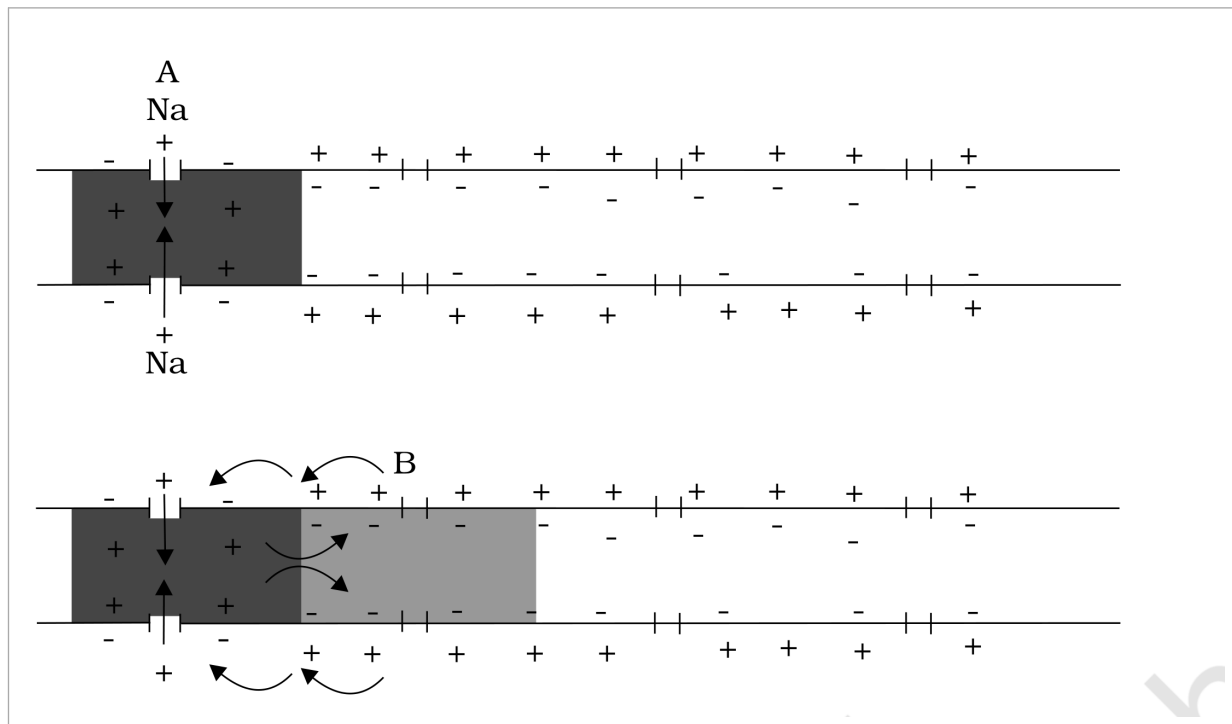


Fig. 18.2 - Diagrammatic representation of impulse conduction through an axon (at points A and B)

- ① [NOTE] Read the plate (Figure 18.2 labels). The plate marks the stimulated site **A** and the site immediately ahead, **B**, with rows of plus and minus charge marks on the two membrane surfaces and **Na** printed at each site where sodium (Na^+) rushes inwards. Reading the panels top to bottom shows the reversed polarity travelling from **A** to **B** along the axon.

Restoring the resting state. The stimulus-induced rise in permeability to Na^+ is extremely short-lived. It is quickly followed by a rise in permeability to K^+ . Within a fraction of a second, K^+ diffuses outside the membrane and restores the resting potential of the membrane at the site of excitation, and the fibre becomes once more responsive to further stimulation.

- ① [NOTE] The impulse travels as a wave. Taken end to end, the nerve impulse is **conducted along the axon membrane in the form of a wave of depolarisation and repolarisation**: depolarisation sweeps forward site by site while the sites just behind it are already being repolarised. (NCERT states this only in the chapter summary; it is folded in here, at the mechanism it describes.)

- ① [NOTE] Exercise gap closed here (Exercise 10b: conduction in a myelinated vs an unmyelinated nerve fibre). The chapter defines both fibre types (18.3) but never contrasts how they conduct. In an **unmyelinated** fibre the bare membrane is depolarised at **every successive point**, exactly as the A-to-B sequence above describes, so conduction is **continuous and slower**. In a **myelinated** fibre the **myelin sheath insulates the axon** and the membrane is exposed only at the **nodes of Ranvier**, so the impulse is regenerated **node to node - saltatory (jumping) conduction**, which is **faster** and uses less energy. Beyond this chapter's own sentences - kept because Exercise 10(b) demands the contrast (Rule 2).

18.3.2 Transmission of Impulses

A nerve impulse is **transmitted from one neuron to another through junctions called synapses**.

- **Synapse** - formed by the membranes of a **pre-synaptic neuron** and a **post-synaptic neuron**, which **may or may not be separated by a gap called the synaptic cleft**.

There are **two types of synapses**, namely, **electrical synapses** and **chemical synapses**.

Feature	Electrical synapse	Chemical synapse
Membranes	Membranes of pre- and post-synaptic neurons are in very close proximity	Membranes of the pre- and post-synaptic neurons are separated by a fluid-filled space called the synaptic cleft (Figure 18.3)

Feature	Electrical synapse	Chemical synapse
How the impulse crosses	Electrical current can flow directly from one neuron into the other across these synapses	Chemicals called neurotransmitters are involved in the transmission of impulses
Resemblance	Transmission is very similar to impulse conduction along a single axon	A fresh potential must be generated in the post-synaptic neuron
Speed	Always faster than that across a chemical synapse	Slower
Occurrence	Rare in our system	The usual synapse in our system

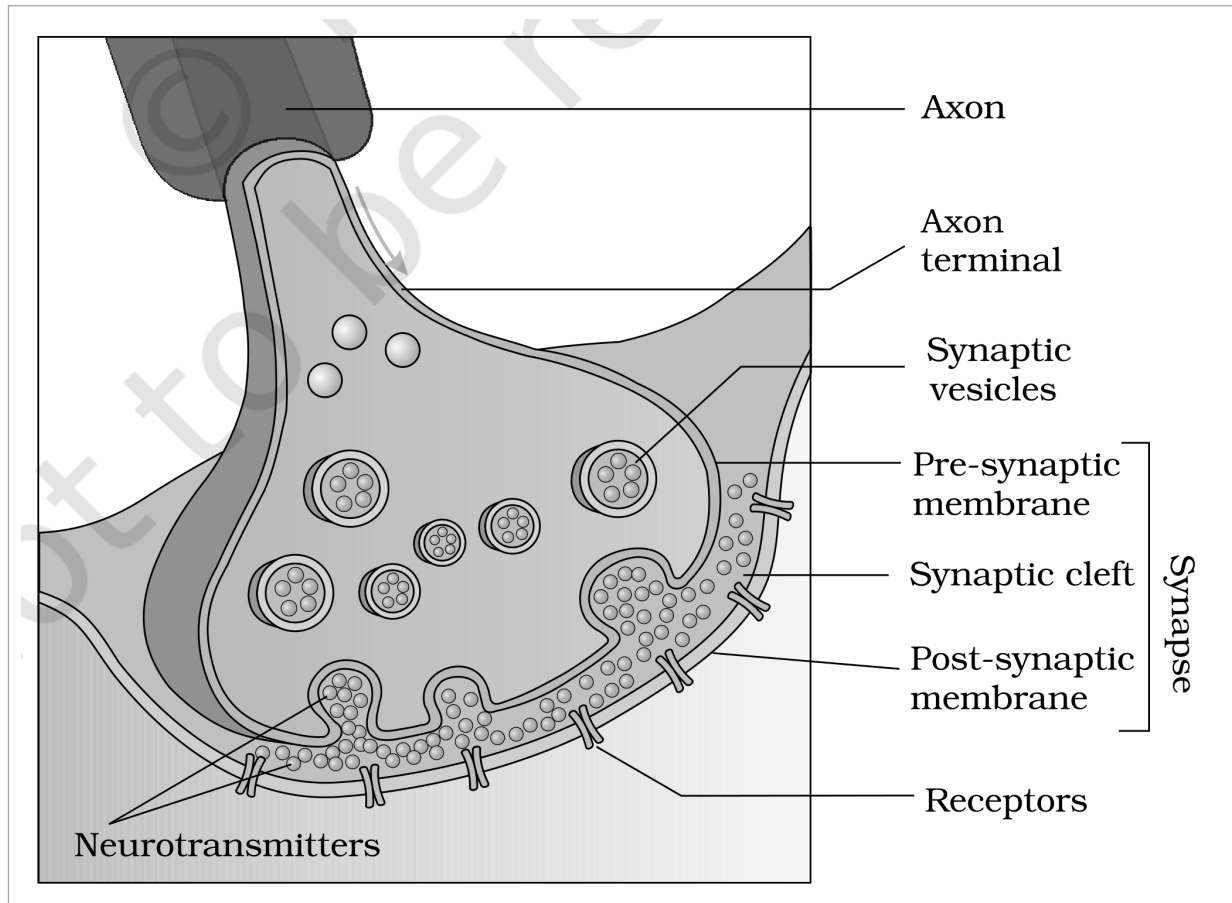


Fig. 18.3 - Diagram showing axon terminal and synapse

① **[NOTE]** Read the plate (Figure 18.3 labels). The plate follows the **Axon** into its **Axon terminal**, which holds the **Synaptic vesicles**; the enlarged inset of the **Synapse** shows the **Pre-synaptic membrane**, the **Synaptic cleft** and the **Post-synaptic membrane**, with **Neurotransmitters** crossing the cleft to the **Receptors** drawn on the post-synaptic membrane.

How does the pre-synaptic neuron transmit an impulse (action potential) across the synaptic cleft to the post-synaptic neuron?

- 1 The **axon terminals** contain **vesicles filled with neurotransmitters**.
- 2 When an **impulse (action potential)** arrives at the **axon terminal**, it **stimulates the movement of the synaptic vesicles towards the membrane**.
- 3 The vesicles **fuse with the plasma membrane** and **release their neurotransmitters in the synaptic cleft**.
- 4 The **released neurotransmitters bind to their specific receptors**, present on the **post-synaptic membrane**.
- 5 This **binding opens ion channels**, allowing the **entry of ions** which can **generate a new potential in the post-synaptic neuron**.
- 6 The **new potential developed** may be either **excitatory or inhibitory**.

The **brain** is the **central information processing organ** of our body, and acts as the '**command and control system**'.

What the brain does	Items NCERT lists
Controls	Voluntary movements; balance of the body; functioning of vital involuntary organs (e.g., lungs, heart, kidneys, etc.); thermoregulation; hunger and thirst; circadian (24-hour) rhythms of our body; activities of several endocrine glands ; and human behaviour
Site for processing of	Vision, hearing, speech, memory, intelligence, emotions and thoughts

Protection. The human brain is **well protected by the skull**. Inside the skull the brain is covered by **cranial meninges**.

Layer of the cranial meninges	Position
Dura mater	The outer layer
Arachnoid	A very thin middle layer
Pia mater	The inner layer, in contact with the brain tissue

The brain can be divided into **three major parts: (i) forebrain, (ii) midbrain and (iii) hindbrain** (Figure 18.4).

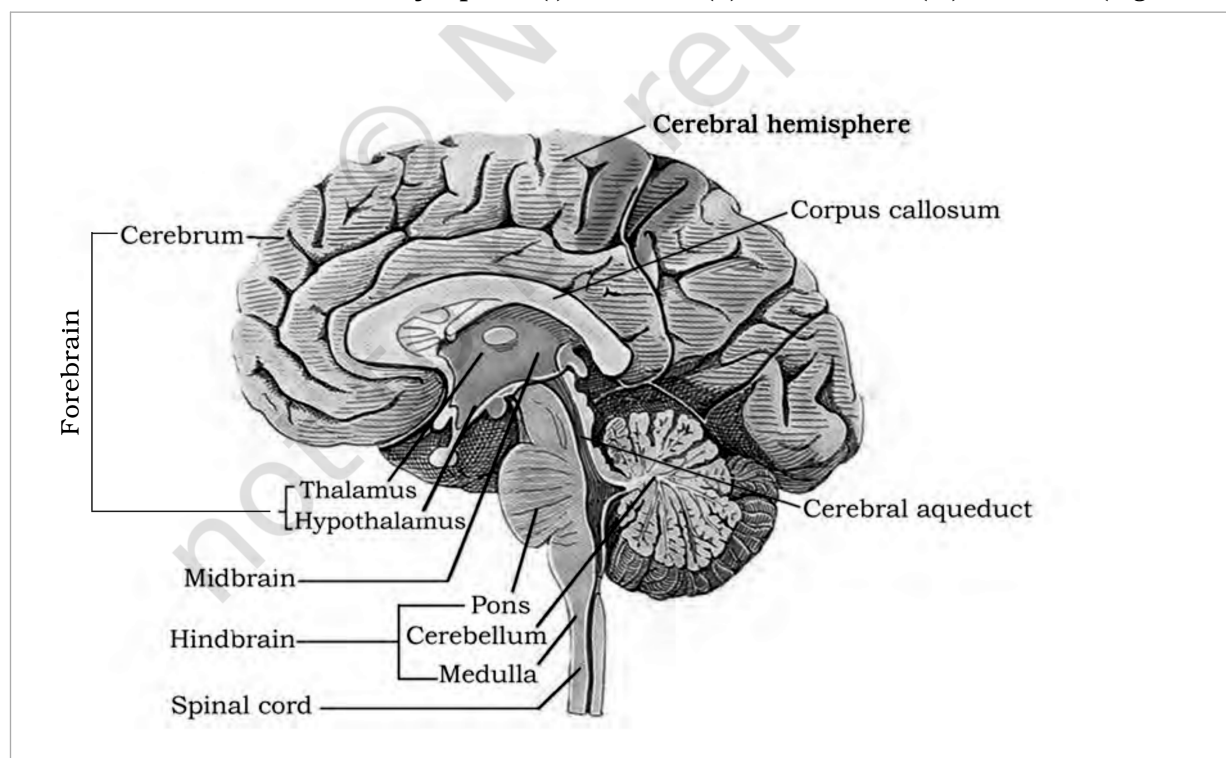


Fig. 18.4 - Diagram showing sagittal section of the human brain (NCERT prints 'sagital'; the conventional spelling is 'sagittal')

① **[NOTE] Read the plate (Figure 18.4 labels).** A bracket marks the **Forebrain**, inside which the plate names the **Cerebrum**, one **Cerebral hemisphere** of the pair, the **Corpus callosum** joining them, the **Thalamus** and the **Hypothalamus**. Below it are the **Midbrain**, with its **Cerebral aqueduct**, and the **Hindbrain** - **Pons, Cerebellum** and **Medulla** - which continues into the **Spinal cord**.

The forebrain consists of **cerebrum, thalamus and hypothalamus** (Figure 18.4).

- **Cerebrum forms the major part of the human brain.**
- **A deep cleft divides the cerebrum longitudinally into two halves, which are termed the left and right cerebral hemispheres.**
- **Corpus callosum** - the **tract of nerve fibres** that **connects the hemispheres**.
- **Cerebral cortex** - the **layer of cells which covers the cerebral hemisphere**; it is **thrown into prominent folds**.

Matter	What it is	Why it looks that way
Grey matter	The cerebral cortex is referred to as the grey matter	Due to its greyish appearance - the neuron cell bodies are concentrated here , giving the colour
White matter	Fibres of the tracts , covered with the myelin sheath , which constitute the inner part of the cerebral hemisphere	They give an opaque white appearance to the layer

The cerebral cortex contains **motor areas**, **sensory areas** and **large regions that are neither clearly sensory nor motor in function**.

- **Association areas** - those neither-sensory-nor-motor regions, responsible for **complex functions** like **intersensory associations**, **memory** and **communication**.

Forebrain part	Position	Function
Thalamus	The cerebrum wraps around it	A major coordinating centre for sensory and motor signaling
Hypothalamus	Lies at the base of the thalamus	Contains a number of centres which control body temperature, urge for eating and drinking ; also contains several groups of neurosecretory cells , which secrete hormones called hypothalamic hormones

- **Limbic lobe or limbic system** - the **inner parts of cerebral hemispheres** and a group of **associated deep structures** like **amygdala**, **hippocampus**, etc., form this complex structure.
- **Along with the hypothalamus**, the limbic system is involved in the **regulation of sexual behaviour**, the **expression of emotional reactions** (e.g., **excitement**, **pleasure**, **rage and fear**) and **motivation**.
- The limbic system is **also concerned with olfaction and autonomic responses**. (*Stated only in the NCERT summary; folded in here with the rest of the limbic functions.*)

① **[NOTE] Exercise gap closed here (Exercise 9b: which part of our central neural system acts as a master clock?).**
The chapter states that the brain controls the **circadian (24-hour) rhythms** of our body (18.4) but never names the timekeeper. That role belongs to the **hypothalamus**, whose **suprachiasmatic nucleus** is the body's **master clock** - consistent with the hypothalamus already being the centre for body temperature, eating and drinking. The name of the nucleus is beyond this chapter's own sentences - kept because Exercise 9(b) demands it (Rule 2).

18.4.2

Midbrain

- The midbrain is **located between the thalamus/hypothalamus of the forebrain and the pons of the hindbrain**.
- A canal called the **cerebral aqueduct** passes through the midbrain (NCERT prints 'passess').
- The **dorsal portion** of the midbrain consists **mainly of four round swellings (lobes)** called **corpora quadrigemina**.
- **Function:** the midbrain **receives and integrates visual, tactile and auditory inputs**. (*Stated only in the NCERT summary; the body gives the midbrain no function at all, so it is folded in here.*)

18.4.3

Hindbrain

The hindbrain comprises **pons**, **cerebellum** and **medulla** (also called the **medulla oblongata**).

Hindbrain part	Structure	Function
Pons	Consists of fibre tracts	They interconnect different regions of the brain
Cerebellum	Has a very convoluted surface in order to provide additional space for many more neurons	Integrates information received from the semicircular canals of the ear and the auditory system (<i>summary-only fact, folded in here</i>)
Medulla	Connected to the spinal cord	Contains centres which control respiration, cardiovascular reflexes and gastric secretions

- **Brain stem** - **three major regions make up the brain stem: mid brain, pons and medulla oblongata**. The brain stem **forms the connections between the brain and spinal cord**.

☆ **[MEMORY AID - not in NCERT] Hindbrain trio - 'PCM':** Pons (bridges brain regions), Cerebellum (balance-and-hearing integrator, convoluted surface), Medulla (respiration, cardiovascular reflexes, gastric secretions). Drop the cerebellum and the remaining two, with the midbrain, are exactly the brain stem.

Recap

Quick Recap (NCERT Summary, rewritten)

The **neural system coordinates and integrates functions** as well as **metabolic and homeostatic activities of all the organs**. Neurons, the **functional units** of the neural system, are **excitable cells** due to a **differential concentration gradient of ions across the membrane**. The electrical potential difference across the **resting** neural membrane is the '**resting potential**'; the nerve impulse is conducted along the axon membrane as a **wave of depolarisation and repolarisation**.

A **synapse** is formed by the membranes of a **pre-synaptic** and a **post-synaptic** neuron, which **may or may not** be separated by a gap called the **synaptic cleft**. Chemicals involved in the transmission of impulses at **chemical synapses** are called **neurotransmitters**.

The human neural system consists of **two parts**: (i) the **central neural system (CNS)** and (ii) the **peripheral neural system**. The **CNS consists of the brain and spinal cord** (the NCERT summary prints 'spiral cord'). The brain divides into (i) **forebrain**, (ii) **midbrain** and (iii) **hindbrain**.

The **forebrain** consists of **cerebrum, thalamus and hypothalamus**. The **cerebrum is longitudinally divided into two halves** connected by the **corpus callosum**. The **hypothalamus, a very important part of the forebrain, controls body temperature, eating and drinking**. The **inner parts of the cerebral hemispheres plus associated deep structures form the limbic system**, concerned with **olfaction, autonomic responses, regulation of sexual behaviour, expression of emotional reactions and motivation**.

The **midbrain receives and integrates visual, tactile and auditory inputs**. The **hindbrain comprises pons, cerebellum and medulla**. The **cerebellum integrates information received from the semicircular canals of the ear and the auditory system**. The **medulla contains centres which control respiration, cardiovascular reflexes and gastric secretions**, and the **pons consists of fibre tracts that interconnect different regions of the brain**.

Ex

EXERCISES (with worked answers)

① **[NOTE]** All **ten** NCERT exercises are reproduced below with worked answers. Every answer is sourced from the chapter text above; the three answers that go beyond NCERT's own sentences (**9b master clock, 10b saltatory conduction, 10f cranial vs spinal nerves**) are flagged in place and collected in the appendix. **Exercise 10 is lettered (a), (b), (f) in the source** - the printed book skips (c)-(e), and that is reproduced, not renumbered.

1. Briefly describe the structure of the Brain.

- The brain is the **central information processing organ**, the '**command and control system**', **protected by the skull** and wrapped in **cranial meninges** - **dura mater** (outer), **arachnoid** (very thin middle) and **pia mater** (inner, touching the brain tissue).
- **Forebrain - cerebrum** (major part; a deep cleft divides it into left and right **cerebral hemispheres** joined by the **corpus callosum**; the folded **cerebral cortex** is the **grey matter** with motor, sensory and **association areas**, over myelinated tracts forming the **white matter**), **thalamus** (coordinating centre for sensory and motor signaling) and **hypothalamus** (temperature, eating, drinking, **hypothalamic hormones**); the **limbic system** lies inside.
- **Midbrain** - between thalamus/hypothalamus and pons; the **cerebral aqueduct** runs through it; the dorsal portion bears the **corpora quadrigemina** (four round swellings) and it integrates **visual, tactile and auditory** inputs.
- **Hindbrain - pons** (fibre tracts interconnecting brain regions), **cerebellum** (very convoluted surface; integrates semicircular-canal and auditory information) and **medulla oblongata** (respiration, cardiovascular reflexes, gastric secretions), connected to the **spinal cord**. **Mid brain + pons + medulla oblongata = brain stem**, the connection between brain and spinal cord (see Figure 18.4).

2. Compare the following:

(a) **Central neural system (CNS) and Peripheral neural system (PNS)**

Feature	CNS	PNS
Parts	Brain and spinal cord	All the nerves of the body associated with the CNS
Role	Site of information processing and control	Carries impulses to and from the CNS through afferent and efferent fibres
Divisions	Forebrain, midbrain, hindbrain; spinal cord	Somatic neural system and autonomic neural system (the latter sympathetic + parasympathetic); the visceral nervous system is the part serving the viscera

(b) Resting potential and action potential

Feature	Resting potential	Action potential
Definition	Electrical potential difference across the resting plasma membrane	Electrical potential difference across the membrane at the stimulated site - in fact termed a nerve impulse
State of the membrane	Polarised	Depolarised (polarity reversed)
Charges	Outer surface positive , inner surface negative	Outer surface negative , inner side positive
Permeability	More permeable to K^+ , nearly impermeable to Na^+ ; gradients held by the sodium-potassium pump (3 Na^+ out for 2 K^+ in)	Membrane becomes freely permeable to Na^+ , giving a rapid influx of Na^+

3. Explain the following processes:

- **(a) Polarisation of the membrane of a nerve fibre.** At rest the axonal membrane is **comparatively more permeable to K^+ , nearly impermeable to Na^+ and impermeable to the negatively charged proteins** of the axoplasm, so the axoplasm holds **high K^+ + negative proteins** and **low Na^+** while the fluid outside holds **low K^+ and high Na^+** - a **concentration gradient** maintained by the **sodium-potassium pump** (3 Na^+ out for 2 K^+ in). Result: **outer surface positive, inner surface negative** - the membrane is **polarised**, and that potential difference is the **resting potential**.
- **(b) Depolarisation of the membrane of a nerve fibre.** A **stimulus** at a site (point A) makes the membrane there **freely permeable to Na^+** ; the **rapid Na^+ influx** reverses the polarity so the **outer surface becomes negative** and the **inner side positive**. The site is **depolarised** and the potential difference there is the **action potential**, i.e. the **nerve impulse**.
- **(c) Transmission of a nerve impulse across a chemical synapse.** See Exercise 6 - the arriving action potential moves **synaptic vesicles** to the membrane, they **fuse** and **release neurotransmitters into the synaptic cleft**, which **bind specific receptors on the post-synaptic membrane**, **opening ion channels** whose **ion entry** generates a new potential that may be **excitatory or inhibitory**.

4. Draw labelled diagrams of the following:

- **(a) Neuron** - redraw **Figure 18.1** and label the **dendrites**, the **Nissl's granules** and **nucleus** in the **cell body**, the **axon** with its **Schwann cell** and **myelin sheath**, a **node of Ranvier**, and the **axon terminal** ending in a **synaptic knob**.
- **(b) Brain** - redraw **Figure 18.4** and label the **forebrain** (**cerebrum**, **cerebral hemisphere**, **corpus callosum**, **thalamus**, **hypothalamus**), the **midbrain** with its **cerebral aqueduct**, and the **hindbrain** (**pons**, **cerebellum**, **medulla**) continuing into the **spinal cord**.

5. Write short notes on the following:

- **(a) Neural coordination** - **coordination** is the process through which two or more organs **interact and complement** each other's functions; the **neural system** does this with an **organised network of point-to-point connections for quick coordination**, while the **endocrine system** provides **chemical integration through hormones**. Example: in exercise, oxygen supply, respiration rate, heart beat and blood flow rise together and return to normal afterwards.
- **(b) Forebrain** - **cerebrum + thalamus + hypothalamus**; cerebrum is the major part, split into **two cerebral hemispheres** joined by the **corpus callosum**, covered by the folded **cerebral cortex** (**grey matter**: motor, sensory and **association areas**) over the myelinated **white matter**; **thalamus** coordinates sensory and motor

signaling; **hypothalamus** controls body temperature and the urge for eating and drinking and secretes **hypothalamic hormones**; the **limbic system** (with **amygdala, hippocampus**) handles **olfaction, autonomic responses, sexual behaviour, emotional reactions and motivation**.

- (c) **Midbrain** - lies **between the thalamus/hypothalamus and the pons**; the **cerebral aqueduct** runs through it; its **dorsal portion** bears **four round swellings, the corpora quadrigemina**; it **receives and integrates visual, tactile and auditory inputs** and, with pons and medulla, forms the **brain stem**.
- (d) **Hindbrain** - **pons** (fibre tracts interconnecting brain regions), **cerebellum** (very **convoluted surface** for more neurons; integrates **semicircular canal** and **auditory** information) and **medulla oblongata** (**respiration, cardiovascular reflexes, gastric secretions**), which is **connected to the spinal cord**.
- (e) **Synapse** - a **junction** formed by the membranes of a **pre-synaptic** and a **post-synaptic** neuron, which **may or may not** be separated by the **synaptic cleft**. **Electrical synapses** (membranes in very close proximity, direct current flow, **always faster, rare in our system**) versus **chemical synapses** (cleft crossed by **neurotransmitters**).

6. Give a brief account of Mechanism of synaptic transmission.

- 1 An **impulse (action potential)** arrives at the **axon terminal** of the **pre-synaptic** neuron, whose **vesicles are filled with neurotransmitters**.
- 2 The impulse **stimulates the movement of the synaptic vesicles towards the membrane**.
- 3 The vesicles **fuse with the plasma membrane** and **release their neurotransmitters into the synaptic cleft**.
- 4 The neurotransmitters **bind their specific receptors on the post-synaptic membrane**.
- 5 The binding **opens ion channels**; the **entering ions generate a new potential** in the post-synaptic neuron, which **may be excitatory or inhibitory**.

7. Explain the role of Na^+ in the generation of action potential.

- At rest the membrane is **nearly impermeable to Na^+** , which is kept **high outside** and **low inside** by the **sodium-potassium pump** (**3 Na^+ out for 2 K^+ in**) - this is what **polarises** the membrane.
- On **stimulation** the membrane at that site becomes **freely permeable to Na^+** ; the **rapid influx of Na^+** **reverses the polarity** (outer negative, inner positive) - **depolarisation** - and the resulting potential difference is the **action potential / nerve impulse**.
- The Na^+ influx also drives the **local current** from the depolarised site to the site immediately ahead, so the impulse is **conducted** along the axon; the rise in Na^+ permeability is **extremely short-lived** and is followed by a rise in K^+ permeability that **restores the resting potential**.

8. Differentiate between:

Pair	First member	Second member
(a) Myelinated vs non-myelinated axons	Enveloped with Schwann cells forming a myelin sheath; gaps between adjacent sheaths are nodes of Ranvier; found in spinal and cranial nerves	Enclosed by a Schwann cell that does not form a myelin sheath; commonly found in autonomic and somatic neural systems
(b) Dendrites vs axons	Short, repeatedly branched fibres projecting from the cell body, containing Nissl's granules; transmit impulses towards the cell body	A long fibre with a branched distal end, each branch ending in a synaptic knob with neurotransmitter vesicles; transmit impulses away from the cell body to a synapse or neuro-muscular junction
(c) Thalamus vs hypothalamus	The cerebrum wraps around it; a major coordinating centre for sensory and motor signaling	Lies at the base of the thalamus; centres for body temperature, eating and drinking; neurosecretory cells secreting hypothalamic hormones; with the limbic system governs emotion, sexual behaviour, motivation
(d) Cerebrum vs cerebellum	Forebrain; major part of the human brain; two hemispheres joined by the corpus callosum; folded cortex with motor, sensory and association areas - voluntary action, memory, intelligence, speech	Hindbrain; very convoluted surface giving space for many more neurons; integrates information from the semicircular canals of the ear and the auditory system

9. Answer the following:

- (a) Which part of the human brain is the most developed? The **cerebrum** - it forms the major part of the human brain.
- (b) Which part of our central neural system acts as a master clock? The **hypothalamus** (its **suprachiasmatic nucleus**), which times the brain-controlled **circadian (24-hour) rhythms**. (Answer beyond NCERT's own sentences - see the appendix.)

10. Distinguish between: (source lettering: (a), (b), (f) - the book skips (c)-(e))

- (a) **Afferent neurons vs efferent neurons.** Afferent nerve fibres **transmit impulses from tissues/organs to the CNS**; efferent fibres **transmit regulatory impulses from the CNS to the concerned peripheral tissues/organs**.
- (b) **Impulse conduction in a myelinated vs an unmyelinated nerve fibre.** In an **unmyelinated** fibre the membrane is depolarised at **every successive point**, so conduction is **continuous and slower**; in a **myelinated** fibre the **myelin sheath insulates** the axon and the membrane is exposed only at the **nodes of Ranvier**, so the impulse leaps **node to node - saltatory conduction, faster** and more energy-efficient. (Answer beyond NCERT's own sentences - see the appendix.)
- (f) **Cranial nerves vs spinal nerves.** Both are **PNS** nerves, and NCERT notes that **myelinated nerve fibres are found in spinal and cranial nerves**. **Cranial nerves arise from the brain**; **spinal nerves arise from the spinal cord**. (Answer beyond NCERT's own sentences - see the appendix.)

Appendix

Terms used in the exercises

Three exercises lean on something this chapter never states. Each is answered above at the point where it belongs, and collected here so the addition is never mistaken for NCERT body text.

Exercise	Term the exercise assumes	Explanation added (not in the NCERT body)
9(b)	Master clock of the central neural system	NCERT states only that the brain controls circadian (24-hour) rhythms . The timekeeper is the hypothalamus , specifically its suprachiasmatic nucleus .
10(b)	Conduction in a myelinated vs an unmyelinated fibre	NCERT defines both fibre types but not their conduction. Unmyelinated: continuous depolarisation point by point, slower . Myelinated: saltatory - the impulse is regenerated only at the nodes of Ranvier , so it is faster .
10(f)	Cranial nerves vs spinal nerves	NCERT names them once, as the site of myelinated fibres. Cranial nerves leave the brain ; spinal nerves leave the spinal cord ; both belong to the PNS and may carry afferent fibres, efferent fibres or both.

- ① **[NOTE] Coverage note.** All 4 figures of the chapter are embedded as verified monochrome assets (Figures 18.1-18.4), each followed by its full in-figure label list, so no figure required manual attention. Source spellings kept verbatim where NCERT prints them: '**sagital**' (Figure 18.4 caption), '**passess**' (18.4.2) and '**spiral cord**' (summary). Nothing in the chapter is a photograph of a person, so no such image was embedded.